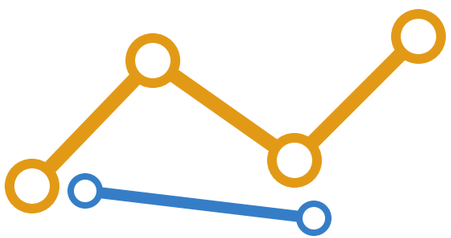
Four-step thesis workflow

from physical model to matched evidence

1 MODEL

Machine physics



Physics prior

articulation + hydraulic contract

Chapter 3

2 LEARN

Actuator dynamics

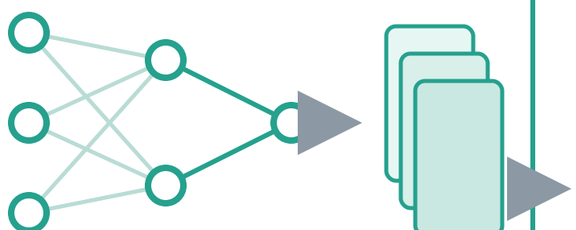
Identified dynamics

Simscape data to compact models

Chapters 4–5

3 TRAIN

Control policy



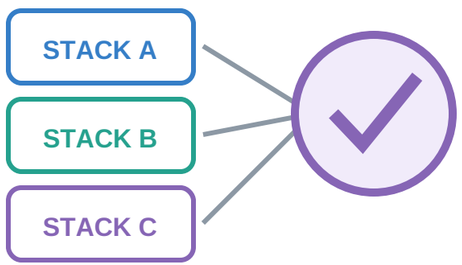
Anchored policy

randomized parallel plants

Chapter 6

4 EVALUATE

Matched stacks



Matched evidence

tracking · robustness · adaptation

Chapter 7

One shared cylinder-motion contract across all stages